

Reentry of Hayabusa Sample Return Capsule and Post-Flight Analysis of the Recovered Heatshield

Tetsuya Yamada¹, Yoshifumi Inatani² and Nobuaki Ishii³
Japan Aerospace Exploration Agency, Sagami-hara, Kanagawa, 252-5210, JAPAN

Ken'ich Hirai⁴, Shinya Morita⁵
IHI Aerospace Co., Ltd., 900 Fujiki, Tomioka-shi, Gunma-ken, 370-2398, JAPAN

Hayabusa sample return capsule (SRC), separated from the mother spacecraft, entered the earth atmosphere over the desert of the Australia on June 13, 2010, and landed safe on the ground after passing through the excessively high aerodynamic heat load with about 14 MW/m². All the component modules of the SRC have been recovered by June 15. The present paper overviews the reentry operation and flight of the SRC together with post-flight analysis of the recovered heatshields: The post-flight analysis program is planned by 3 steps and is under progress : The first step is reconstruction of the reentry flight trajectory and the flight environment. The reentry trajectory is to be reproduced synthetically taking account of the accuracy of the reentry orbit determination, the SRC landing point, atmospheric density and the wind on the reentry day. And the second step is to overview the SRC through non-intrusive method such as X-ray CT-scanning. The 3-dimensional laser-scanning has been also carried out for measuring the surface recession. Regardless of difficulty in the surface roughness, emissivity dispersion etc., 3-D surface has been numerically reconstructed and compared with the CAD data at the pre-flight.

Nomenclature

LAC	= linear absorption coefficient of X-ray
w	= weight fraction
τ	= absorption for an element at a specific energy of the X-ray
ρ	= density of the ablator

I. Introduction

THE Asteroid Explorer Hayabusa was launched by M-V rocket from Uchinoura Space Center, ISAS/JAXA on May 9, 2003. Conquering several troubles encountered during totally 7 years of orbital flight, it returned to the vicinity of the earth and completed the powered-flight by the ion thrusters in the beginning of 2010. After successive trajectory correction maneuvers for the reentry, the mother spacecraft successfully separated a small sample return capsule (SRC) with 40cm in diameter accommodating asteroid sample in it and the SRC has entered the earth atmosphere over the Woomera desert in the South Australia on June 13, 2010. The SRC successfully landed on the ground after passing through the excessively high aerodynamic heat load with about 14 MW/m². All the component modules of the SRC have been recovered by June 15 [1]. Successful recovery of the reentry capsule is since USERS REM capsule in 2003 [2], and Hayabusa SRC is the first capsule in Japan which accomplished direct reentry from the interplanetary transfer orbit. Design of the Hayabusa SRC started in 1995 [3], and successful recovery provided us of precious opportunity for analyzing high-speed reentry flight and SRC itself after undergoing 7 years' exposure to space environment. Synthetic analysis process for the SRC has started aiming the completion by the end of 2011

¹ Associate Professor, Div. Space Systems and Astronautics.

² Professor, Div. Space Systems and Astronautics.

³ Professor, Div. Space Systems and Astronautics.

⁴ Technology Manager, Space Development Section.

⁵ Technology Manager, Space Development Section.

fiscal year. The present paper overviews the reentry flight of the Hayabusa SRC and post-flight analysis of the recovered heatshields.

II. Reentry Flight of the Sample Return Capsule

A. Reentry Parameter Setting

According to accuracy of the orbital determination [4], 4 major trajectory correction maneuvers (TCM) were carried out. A series of preparatory operations for SRC were executed before the actual reentry:

Temperature of the onboard instrument needs to be maintained beyond -20°C for normal reentry sequence. The temperature of the SRC is estimated to decrease gradually during its sole flight after separated from the mother spacecraft and its battery capacity is not enough to compensate its temperature decrease within appropriate range. Thus, the SRC needs to be warmed-up to appropriate temperature potential by mother spacecraft so that the temperature decrease of the SRC during its sole flight is made minimum. Consequently, the maximum allowable temperature of the SRC was set to 5°C because it was anticipated that the warm-up beyond it might cause vaporization of the RCS propellant that might have been frozen and attached to the body in the trouble encountered on the Itokawa. SRC was, thus, separated 3 hours before the reentry as the time sequence is shown in Table -1.

SRC enters the earth atmosphere and after passing through the aerodynamic heating corridor, it has jettisoned the aftbody and the forebody heatshield at the altitude of about 5 km. The instrument module (IM : the main body of the SRC including the sampler container) gradually made decent with parachute, and it started beacon signal transmission and the heatshield have made impact landing on the ground. SRC adopts two triggering methods. The primary is a G-sensing method, in which the chute triggering timer is activated at a given deceleration level (5 G was set as a result), and the chute is deployed after a given seconds (244 sec) after trigger generation. The secondary timer, prescribing the time after the separation, behaves a backup at the estimated latest time in case that chute deployment is not normally triggered. Malfunction of the accelerometer leads to fatal incident, the timer setting was investigated to minimize degree of

Table-1. Reentry Time Sequence

Time [UT]	Event
6:51	- SRC Power On by external power - write reentry parameters
7:21	- Switch from Ext. to Int. Battery
7:51	- Sequence Start
10:45	- interface cable cut
10:51	- SRC Separation
13:51:07	- Accelerometer offset measurement
13:51:11	- SRC @ H=200km
13:52:19	- G-triggered (@5G)
13:56:23	- Parachute deployment
13:56:31	- Beacon Transmission Start
about 14:09	- landing (nominal prediction)
14:22	- Chute-anchor Separation
17:55	- Beacon stop

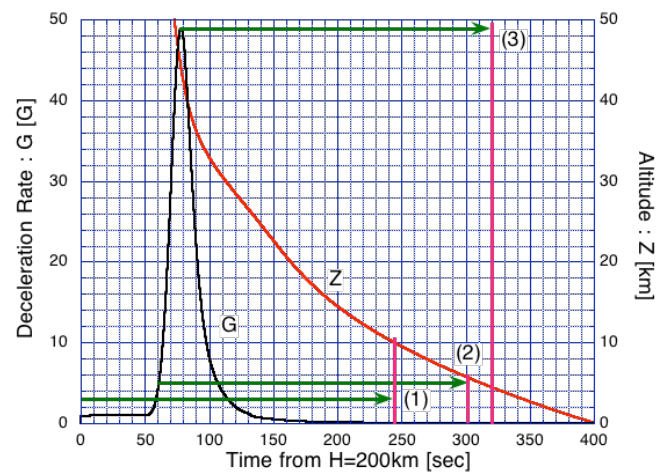


Fig.1 Deceleration and the cases of parachute Triggering Cases at 1) at the time of off-set measurement. 2) at the nominally -sensed deceleration 3) at the maximum deceleration.

chute-deployment failure mode: By setting 5 km as the nominal chute-deployment altitude, SRC deploy the chute between 10 km to 5 km corresponding trigger signal generated at accelerometer switch-on to at 50 G. The time of event including reentry parameter setting is shown in Table 2. Actually the accelerometer was considered to function at 5G as planned and the SRC is estimated to deploy chute at the altitude about 5 km.

B. Reentry Flight and Recovery Operation

Both of the forebody and the aftbody heatshields are separated from the instrument module simultaneously by one action of the pyrotechnic chute-separation mechanism. Four seconds after the chute deployment, SRC starts transmitting beacon signal. During slow descent phase of the IM with chute, localization is carried out based on the beacons signal received by DFS' as a cross point of each azimuth direction vector from a DFS [5]. At the (estimated maximum) landing time of SRC, the helicopter hovering in the air outside a given impact probability ellipse starts to target the position indicated by DFS' or by its onboard direction finding system. The beacon signal from the IM has been easily detected during its descent by 4 direction finding stations located around the landing dispersion. According to the azimuth direction information, landing point of the IM was analyzed at the JAXA onsite HQ located in the range control center (RCC) in Woomera Range Head. The helicopter (S-74), under the contract between JAXA and the AOSG/RAAF (Aerospace Operational Support Group/ Royal Australian Air Force) has launched and made flight to the estimated landing point. The IM was discovered about 1 hour after the launch by S-76 with a bright floodlight and the beacon receiver installed. The forebody and aftbody heatshield should have been searched out by an infrared camera mounted on the helicopter during the night before the heatshields experience the daylight. Because the daylight will undesirably equilibrate the heatshields' relatively hot temperatures with the ground temperature, which makes it difficult to discriminate them. Though heatshields were not discovered during the night-time by IR camera, they have been recognized on the video reproduction later. Landing position of the other 2 objects, namely forebody and aftbody heatshields were investigated based on the actual landing point of the IM using the wind data supplied through the website of NCEP (National Centers Environmental Prediction, NOAA): Parachute deployment point was predicted by fitting the most probable descent trajectory so as it land on the actual landing point, and the landing point of heatshields are, in turn, analyzed. Figure

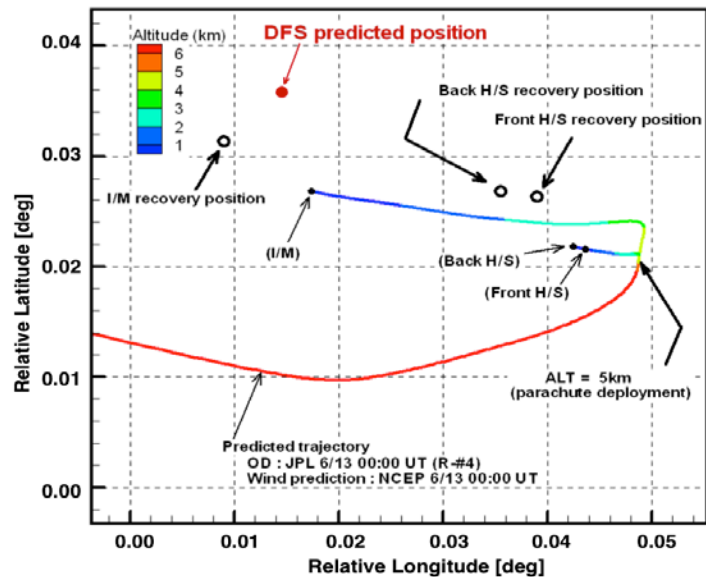


Fig. 2 Recovered 3 components and the positional relations between prediction and actual recovered points.

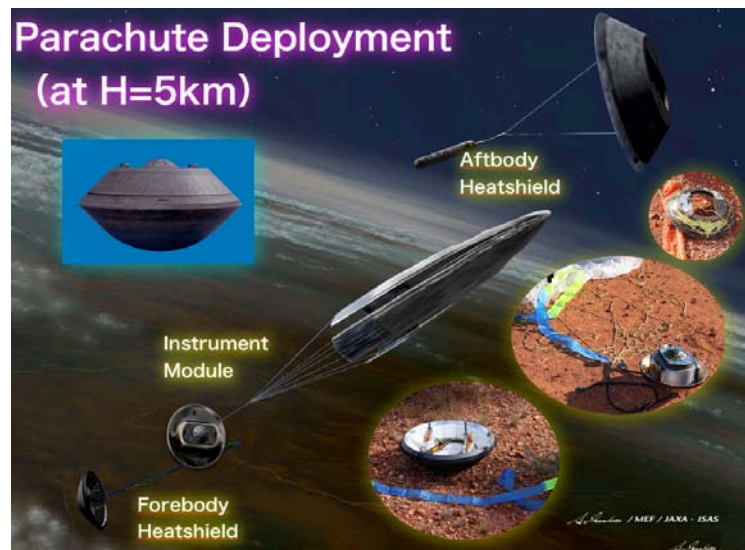


Fig. 3 Artist's Impression for Parachute Deployment and the Recovered 3 Components.

2 shows the location of the recovered points of the scheduled landing objects. The heatshields have been discovered within 200 m from predicted landing point owing to the lucky situation of relatively weak wind on the day. Three scheduled landing objects have been discovered within 12 hours after reentry and recovered within 36 hours by June 15 as schematically shown in Fig. 3.

III. Flight Environment and Post-Flight Analysis

A. Synthetic Analysis Planning of SRC

Successful recovery provided us of precious opportunity for analyzing high-speed reentry flight and SRC itself after undergoing 7 years' exposure to space environment. Synthetic analysis process for the SRC has started aiming completion by the end of 2011 fiscal year.

Items which planned to be analyzed are the strength degradation, permeability characteristics of the parachute, and characteristics of onboard electronics. Residual capacity of the battery has been measured after recovery. Measured capacity residual corresponded to the value calculated based on the actual consumption during the orbital flight and the beacon transmission.

The post-flight analysis of the recovered heatshield, thus, need to be carried out carefully and the inspection goals are 1) to record the dimension and weight of the each heatshield component and assembly as compared with pre-flight condition and 2) to evaluate the integrity of the heatshields from the thermal and mechanical point of view and 3) to provide knowledge to possible future missions which will adopt similar material, through the evaluation of the heatshield material. The post-flight analysis program is planned by 3 steps and is under progress.

B. Estimation of Reentry Flight Environment

The first step in the post-flight analysis is reconstruction of the reentry flight trajectory and the flight environment. Due to the poor weight budget for the SRC, no sensors for the reentry flight state measurement were equipped with onboard. Thus, the reentry trajectory needs to be reproduced synthetically taking account of the accuracy of the reentry orbit determination, the SRC landing point, atmospheric density and the wind on the reentry day.

As described in the previous section, the SRC components were found within the range of 1 km from the prediction based on the nominal trajectory and nominal wind prediction. Because the difference in the landing point

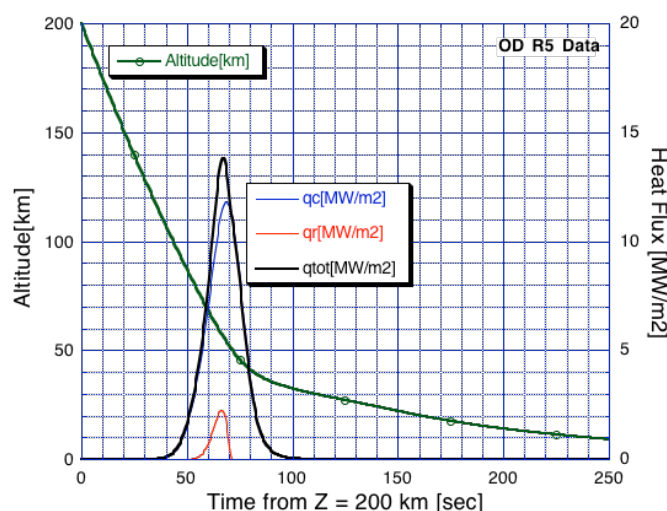


Fig. 4 Time History of the Aerodynamic Heating.(qr/qc : radiative/ convective heatflux)

and prediction is relatively so small that it can be easily explained by a little dispersion of the wind velocity profile from the nominal, it is not reasonable to consider dispersions in orbital entry conditions, atmospheric density, and errors in aerodynamic coefficient as major incident of the landing point difference at the first approximation. Just applying nominal values in entry conditions, atmospheric density based on GRAM-99, and aerodynamic coefficients, the most probable reentry flight environment has been calculated and shown in Fig. 4. Detailed analysis is now under progress based on the optical observation data obtained by the JAXA ground observation team and NASA/JAXA joint airborne fireball observation team onboard DC-8.

C. Fireball Observation by 4-camera with spectroscopic filters

There is not a so many opportunities of the reentry with super-orbital velocity even in the world. In order to study the radiation of the reentering mother spacecraft and the capsule, the radiation observation campaigns were carried out both in the ground and the airborne. In case of contingency that no beacon signal is transmitted at from the SRC and all the ground direction-finding stations failed to localized it, the reproduction of the reentry trajectory is of prime importance. The airborne fireball observation by using DC-8 aircraft was organized based on the agreement between NASA and JAXA. JAXA is responsible for providing necessary report on the progress of the Hayabusa mission and latest orbital information, while NASA is responsible for the operation of DC-8. The science outcomes are to be shared among the Hayabusa Joint Science Team (JST).

In order to separate the radiation signal of the SRC from the those of mothership components under destructive entry, where the distance between them was estimated below 2 km, 4-camera system with relatively broader visual field of 200 mm telescope and optical filters was adopted for JAXA's airborne fireball observation system. The characteristics of optic filters installed on the cameras are as shown in Fig. 5.

After careful simulation of the consequent radiation data when the radiation temperature is fixed by taking account of optical conversion efficiencies, the temperature profile based on the blackbody radiation from the SRC has been dedicated and shown in Fig. 6. Though the temperature obtained here may be an averaged value among surface plasma, radiation from the SRC surface, it proved to be 3,400 K at maximum.

D. Reassembling of SRC

The recovered 3 components were re-assembled to evaluate thermo-mechanical integrity of the heatshield, to survey overall shape distortion and to measure qualitatively the gap between the forebody and aftbody heatshield. Regardless of the impact on the ground at the velocity about 40m/s (forebody heatshield) and 30m/s (aftbody), respectively, no apparent shape distortion was observed and the measured gap between forebody and aftbody heatshield ranged from 1.00 to 1.20 mm in average while the designed gap value is 1.0 mm. As mentioned in the previous section, the heatshields are jettisoned at the altitude of 5km and the forebody/aftbody heatshield impacted on the ground with the terminal velocity of 40m/s and 30 m/s, respectively. Nevertheless, the distortion at the ground impact seems to be severe and the gap between the two

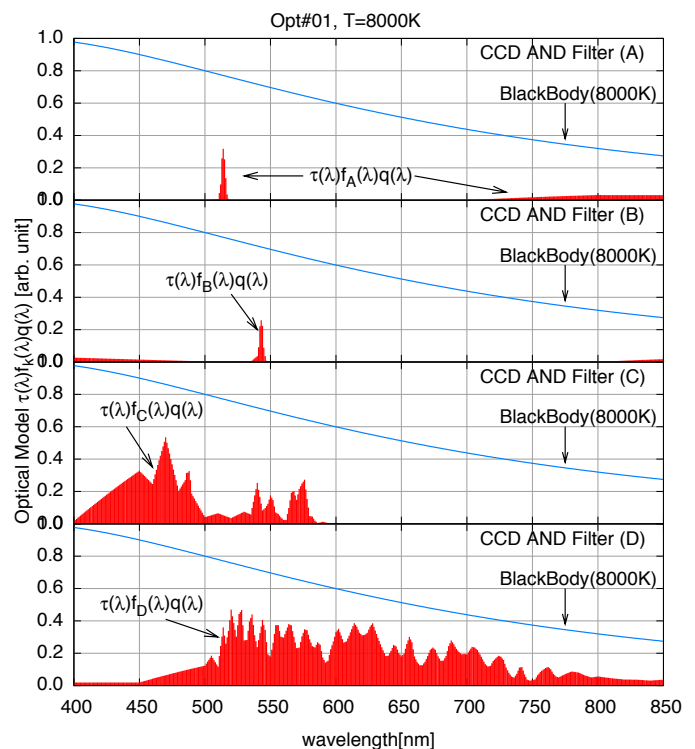


Fig. 5 Characteristics of the each optic filter installed on the JAXA's Hayabusa airborne 4-camera fireball observation system.

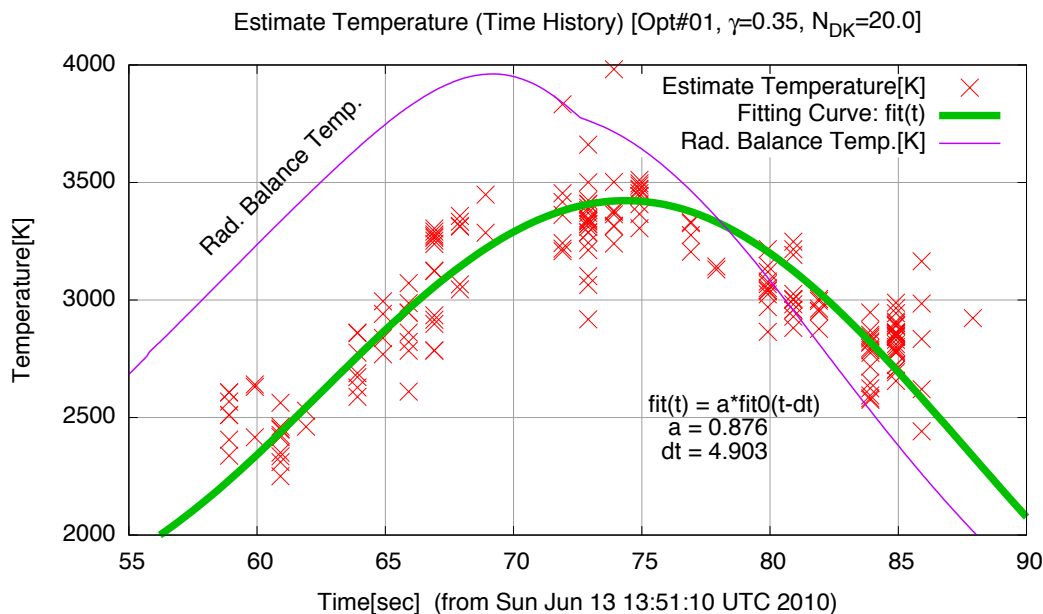


Fig. 6 Temperature Profile estimate based on the radiation from the Hayabusa SRC.

heatshield reassembled remained same as the designed before actual reentry flight. Precise measurement of the overall SRC shape by means of 3-dimensional laser scanning is described in the next section.

VI. Heatshield Inspection

A. Issues on Forebody Heatshield

Estimation of the heatflux distribution around SRC was of prime interest, especially the existence of the heatflux enhancement due to the boundary layer transition from laminar to turbulent. There were several numerical and experimental analysis carried out at the beginning of the design process of the heatshield [for example 9-11]. Because onset of the boundary layer transition was not completely denied, the heatflux distribution around the forebody heatshield was assumed to be flat as shown in Fig. 7. The recovered heatshield was considered to reveal us of the onset of the transition.

Ablator material of the forebody and aftbody heatshield is basically composed of carbon phenolic. In the excessively high aerodynamic heat load environment, the production rate of the pyrolysis gas in the reaction layer is high, and the internal pyrolysis gas pressure sometimes exceeds the allowable stress, which causes ply delamination. In order to avoid the delamination, the lattice-layup method is adopted in the stagnation region in the top sphere section of the heatshield. The lay-up angle to the surface is different according to the portion in the heatshield as shown in Fig. 8, the cloth-layered ablator tilted by 22.5 deg at maximum to the surface is used in conical part. Two ablator components are combined by using adhesive of AREMCO-805, whose decomposition start at about 400°C. The design of the heatshield needs to be validated based on the recovered objects.

B. 3-dimensional Recession Measurement

In order to obtain the overall recession characteristics over the forebody heatshield, 3-dimensional shape measurement has been carried out by 3-dimensional laser scanner (Konica Range-7) with Z-direction (line-of-sight direction) accuracy of 80 μm . Regardless of difficulty in the surface roughness, and emissivity dispersion etc., 3-D surface has numerically reconstructed surface scanning data. One shot picture with size of 10cm by 10cm (1280 x

1024 pixels) was combined or connected based on the marker attached on each picture object to construct overall mapping. A given point at the bottom of the backside of the heatshield was selected as a fixed point or an original point for reconstruction. The reconstructed 3-dimensional data were compared with the CAD data taken before launch and the overall recession was obtained after numerical process. Because the backside shape coincided with the original designed shape, the effect of the overall shape distortion at ground impact need not to be considered, and we can continue discussion on the surface recession; Figure 9 shows 3 dimensional recession mapping of the forebody heatshield; The maximum recession with 0.3 mm has been measured in the vicinity of stagnation region, and on the contrary, the swelling with 0.3 mm at maximum is observed in the conical section. The consequent swelling was not estimated before actual flight: It is true the heatshield ablator material proved to swell somehow at the ground heating tests, but the recession is dominant and the ablator, consequently, recesses and the recession less than 1 mm was actually predicted for reentry.

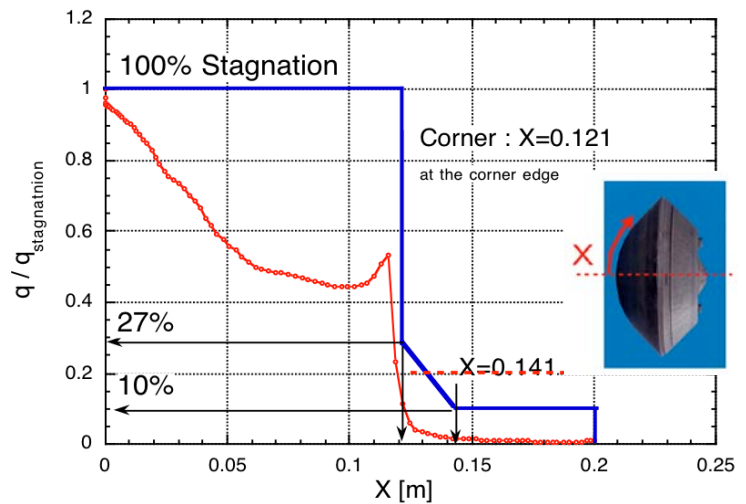


Fig. 7 Heat flux distribution around the SRC (Analytic Prediction and Design Result)

C. X-ray CT Scanning of Forebody Heatshield

The non-intrusive technique such as X-ray CT-scanning [12] is very useful for avoiding unnecessary intrusive or irreversible method such as gouging samples out of the heatshield. The linear absorption coefficient (LAC) of the X-ray is strong function of the density and the element composition of the material. Though the element

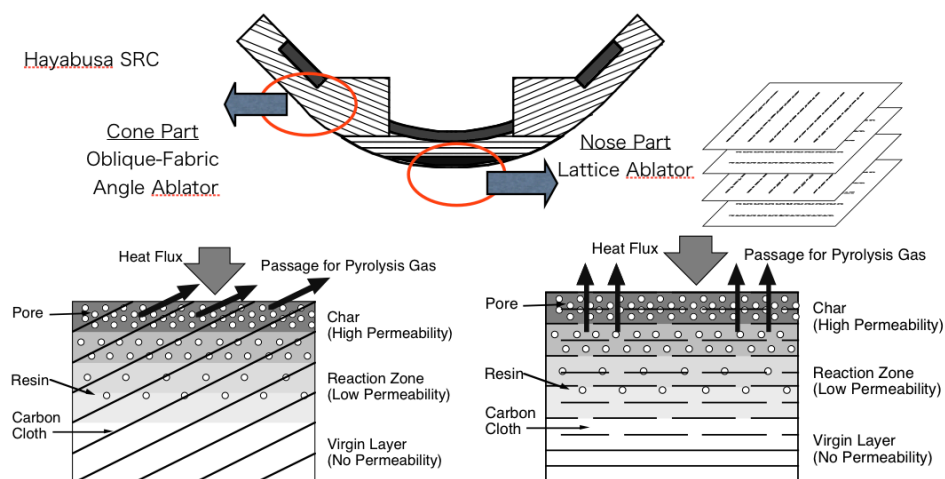


Fig. 8 A Schematics of the Structure of the Forebody Heatshield.

composition of the ablator is complicated, LAC can be correlated to the density of the ablator if the LAC value is well-correlated to the ablator density in advance by using arc heated test piece with a given density.

$$\rho = \frac{LAC}{\sum w_i \cdot \tau_i(E)}$$

There exist several ‘artifact’ that makes it difficult to inspect the image of the material. Among them, large difference in X-ray absorption coefficients of materials produces streak and star pattern artifacts in the CT images. The artifact, which is called "metal artifact", makes it difficult to inspect the image of the device. Narrow-band X-

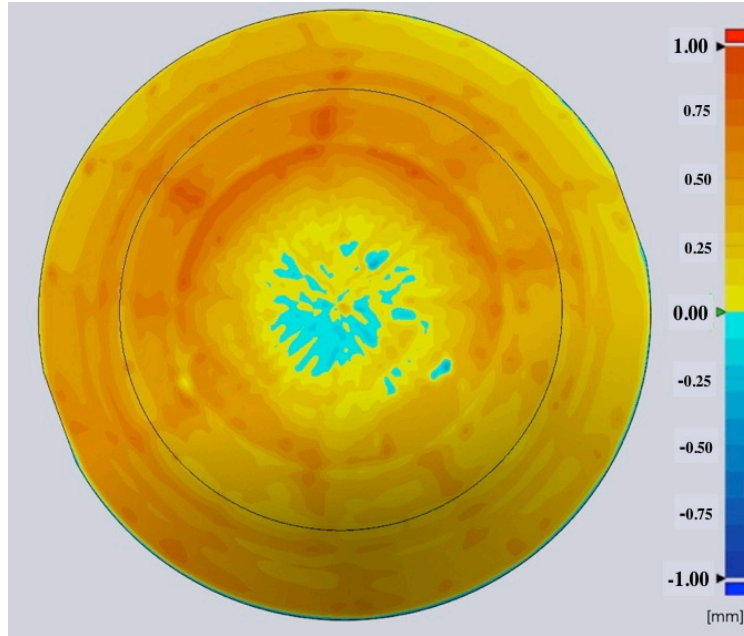


Fig. 9 Surface recession mapping. (the scale is not added at the abstract submission)

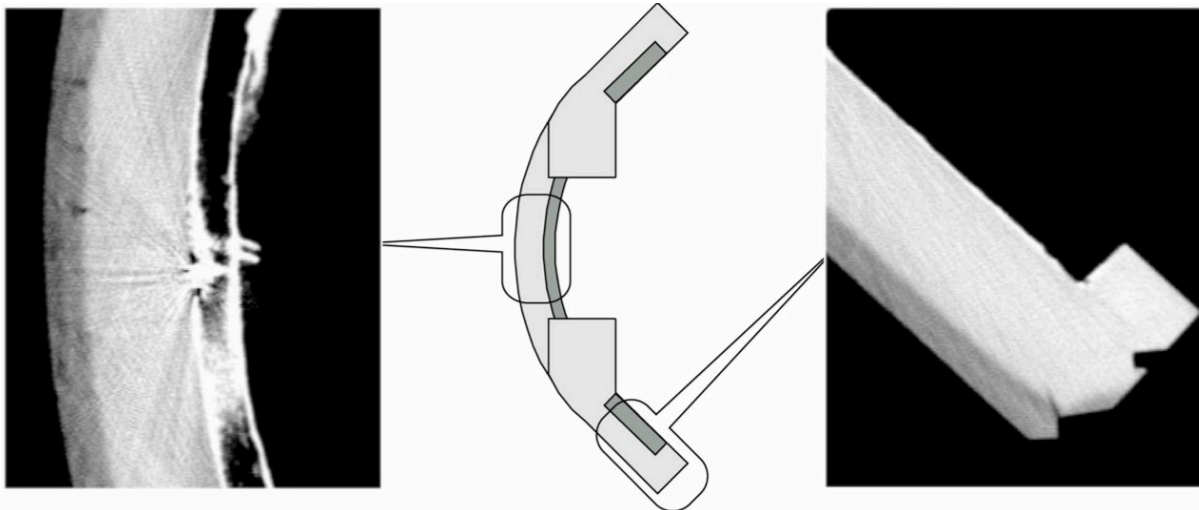


Fig. 10 A raw X-ray image of the forebody heatshield: In the (a) stagnation region (left), the metal-artifact is observed due to fastener-screw to fix backside Titan plate for reinforcing joint region of nose and conical part of the heatshield, while no metal-artifact is observed in (b) conical region.

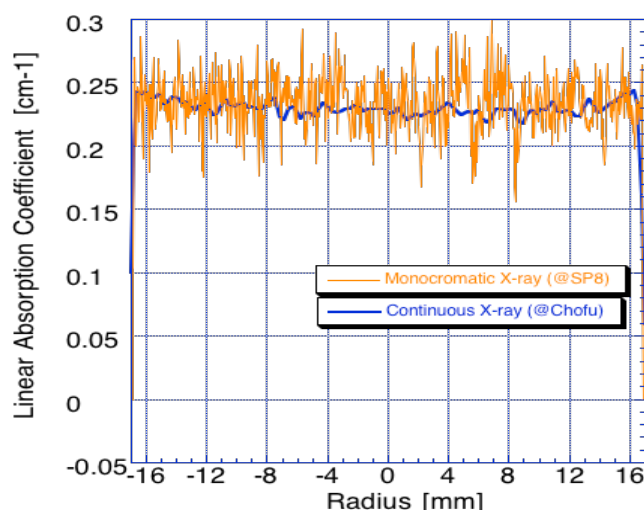


Fig. 11 LAC characteristics along the radius direction of uniform virgin layer of the ablator with 50 mm in diameter; Beam-hardening is not observed.

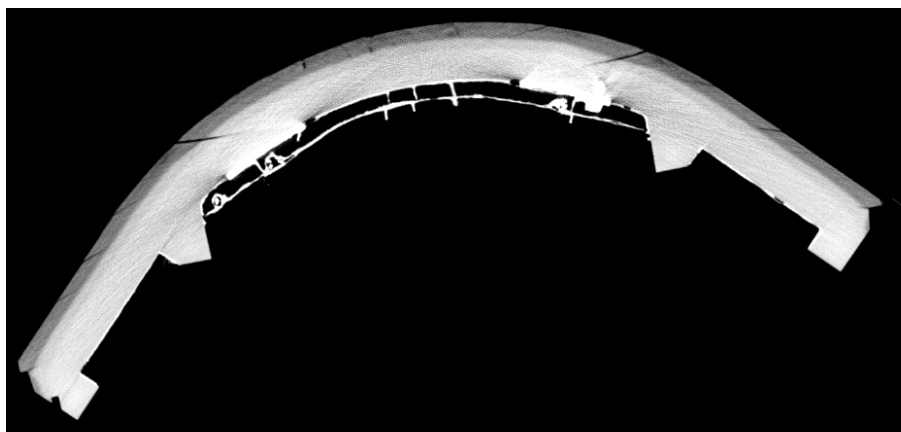


Fig. 12 Typical image of the X-ray CT-scanned forebody heatshield. The char layer and the pyrolysis layer can be distinguished by the color difference. The dark-line between the nose and the conical section is NOT a crack but filled with epoxy adhesive-bond with lower LAC than the phenolic resin.

ray is considered to be advantageous to the broad-band from the standpoint of “beam-hardening” effect. In case that the beam hardening effect can not be ignored, the X-ray absorption characteristics exhibits distribution along depth regardless of the flat density distribution, which deteriorates the validity of principle of density measurement by X-ray tomography.

By adjusting the beam power and the band width, we could successfully obtained the LAC distribution map in the regions of char layer, pyrolysis layer, and virgin layer over the heatshield both in JAXA Composite Material Center (Broad-band X-ray) and SPring-8 (narrow-band X-ray), respectively. Figure 10 shows an example of raw X-ray image of the forebody heatshield: In the (a) stagnation region (left), the metal-artifact is observed due to fastener-screw to fix backside Titan plate for reinforcing joint region of nose and conical part of the heatshield, while no metal-artifact is observed in (b) conical region. Though it is difficult to obtain the images completely free from metal-artifact, we can continue discussion by selecting X-ray cross-section which has no metals included in a given tomographic plane.

The degree of the beam hardening has been measured and estimated: Figure 11 shows LAC characteristics along the radius direction in the uniform virgin layer of the ablator test piece with 50mm in diameter: The measure LAC exhibits almost flat along radial direction though the noisy data of SPring-8 are caused by its excessively excellent resolution smaller than fiber and pore size. Anyway this fact endorses the measurement validity that the beam-hardening effect can be ignored, because the thickness of the recovered heatshield is about 37 mm at maximum.

The measured LAC distribution and the degree of reaction (namely the density difference) has been correlated based on cross-section of the arc-heated test piece and its X-ray imaging. The thickness of the charred layer is almost uniform value of about 10mm all over the forebody heatshield as shown in Fig.12. This fact does not necessarily mean the enhancement of the heatflux due to the boundary layer transition in the conical part, but the difference in the thermal diffusivity needs to be carefully considered for further numerical analysis which now undergoing and preliminary reported in the Ref. [12]. In Fig. 12, we can recognize the dark line in the juncture between the nose part and the conical part, which is not crack. The nose and the conical part are combined with adhesive-bond AREMCO-805 mainly made of an epoxy-type resin. Because the LAC of the epoxy adhesive-bond seems to be between that of the air and the phenolic resin, the contrast of Figure 12 is adjusted to make the boundary between pyrolysis layers. Actual cross-section inspection is planned to be in the later half of 2011.

D. Aftbody Heatshield

Figure 13 shows the recovered aft-body heatshield. It is easily recognized that the surface thermal-control tape (Aluminum-deposit polyimide tape) is partially left on the backside surface. In the design phase, the estimation of the heatflux on the aftbody was of crucial matter. Taking account of the reaction between ablation products flown from the forebody heatshield and the nonequilibrium high enthalpy air, the radiation heatflux enhancement due to the recombination was anticipated in the aft wake region. A lot of research activities were carried out [for example 14-15]. Finally 10% of the stagnation point (namely 1.5 MW/m²) was assumed at maximum heatflux time point as previously shown in Fig. 7. The recovered aftbody heatshield shows us that the heatflux at backside is small enough that the maximum temperature is estimated about 400 °C, which is onset temperature of the decomposition of the polyimide film.



Fig. 13 Aftbody Heatshield and the structural validity of the ventilation holes.

In order to deploy the parachute properly, the internal pressure of the SRC needs to be equilibrated to the atmospheric pressure by 70% at lowest from the standpoint of the pyrotechnique devise energy required for the parachute cover jettison. The ambient air was taken into the inside of SRC during its reentry phase through the small 2 mm holes located at the backside as shown in the magnified photo in Fig. 13. The hot air flow taken in through the orifice holes confronts a metal wall opposite to the holes and the intake path is carefully designed with several corners to enhance heat absorption by the walls. Total heat absorption due to the incoming hot flow to fill the inside to the appropriate pressure level was estimated to be about 500 J during the reentry phase.

Owing to the low heatflux, the size of the hole remained to be same as designed: They were not shrank nor expanded. And no heat damages were found on the inside wall nor in the parachute accommodated inside the capsule.

V. Concluding Remarks

The Hayabusa sample return capsule (SRC), after separated from the mother spacecraft 3 hrs before its reentry, entered the earth atmosphere over the desert of the Australia on June 13, 2010. It landed successfully on the ground after passing through the excessively high aerodynamic heat load with about 14 MW/m². The present paper has overviewed the reentry flight of the Hayabusa SRC and post-flight analysis of the recovered heatshields. The reentry trajectory was reproduced mainly based on the recovered point; Taking account of the accuracy of the reentry orbit determination, the SRC landing point, atmospheric density and the wind on the reentry day, the produced reentry is almost nominal. The SRC was surveyed through non-intrusive method such as 3-dimensional laser displacement scanning and X-ray CT-scanning. Regardless of difficulty in the surface roughness, and emissivity dispersion etc., 3-D surface has been numerically reconstructed and compared with the CAD data at the pre-flight and showed the recession in the stagnation region and swelling in most part with ± 0.3 mm at maximum, respectively. The uniform thickness of the charred layer was observed with X-ray CT scanning of the overall heatshield. All these have endorsed the design validity of the heatshield thickness.

The synthetic post-flight analysis is planned to be completed within 2011 fiscal year, lessons learned will contribute to designing of the further small reentry capsule.

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